



Pangenome and Genomic Signatures of Lineage-4 *Mycobacterium tuberculosis* in Extrapulmonary TB (Western Ethiopia)



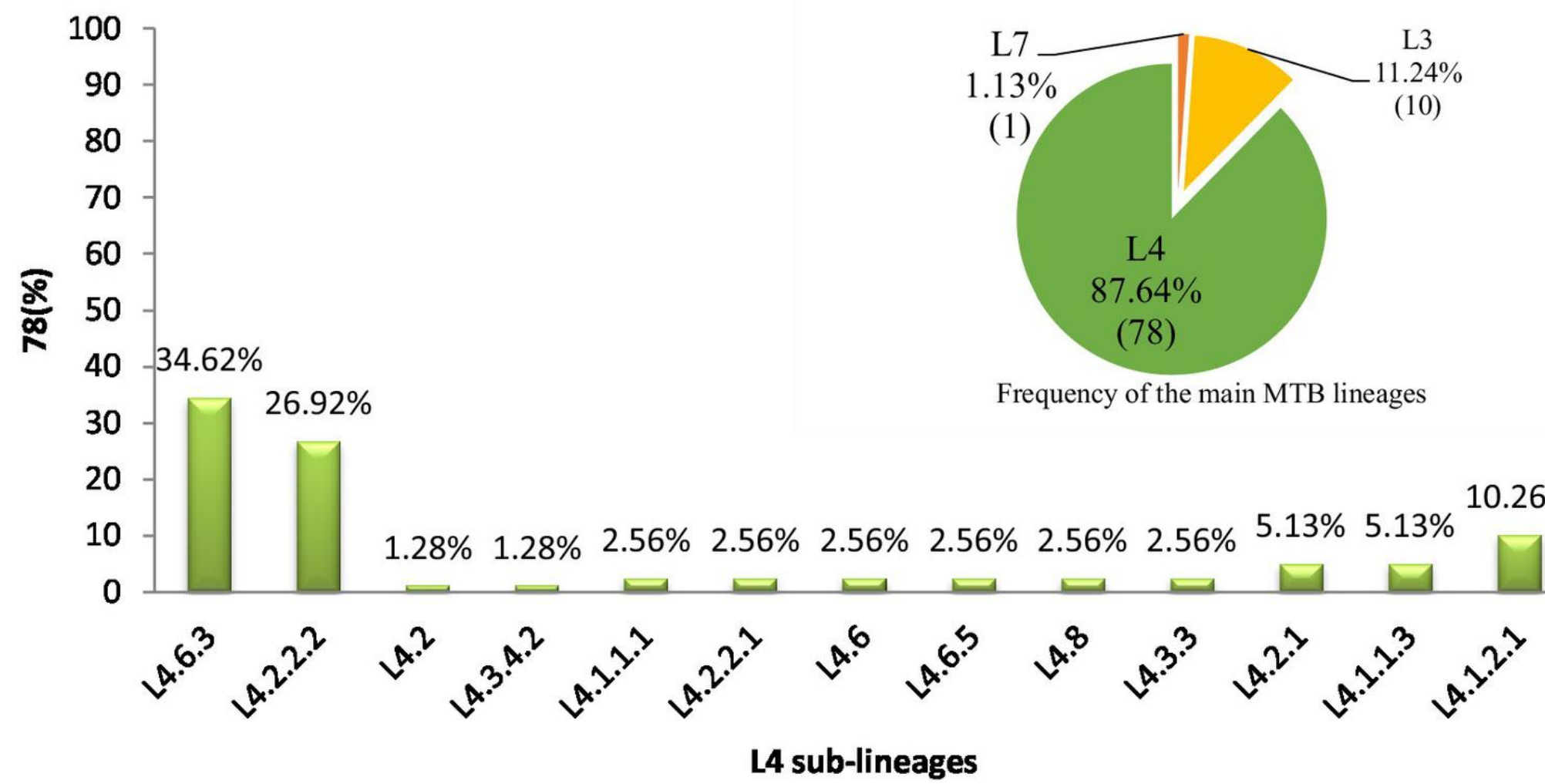
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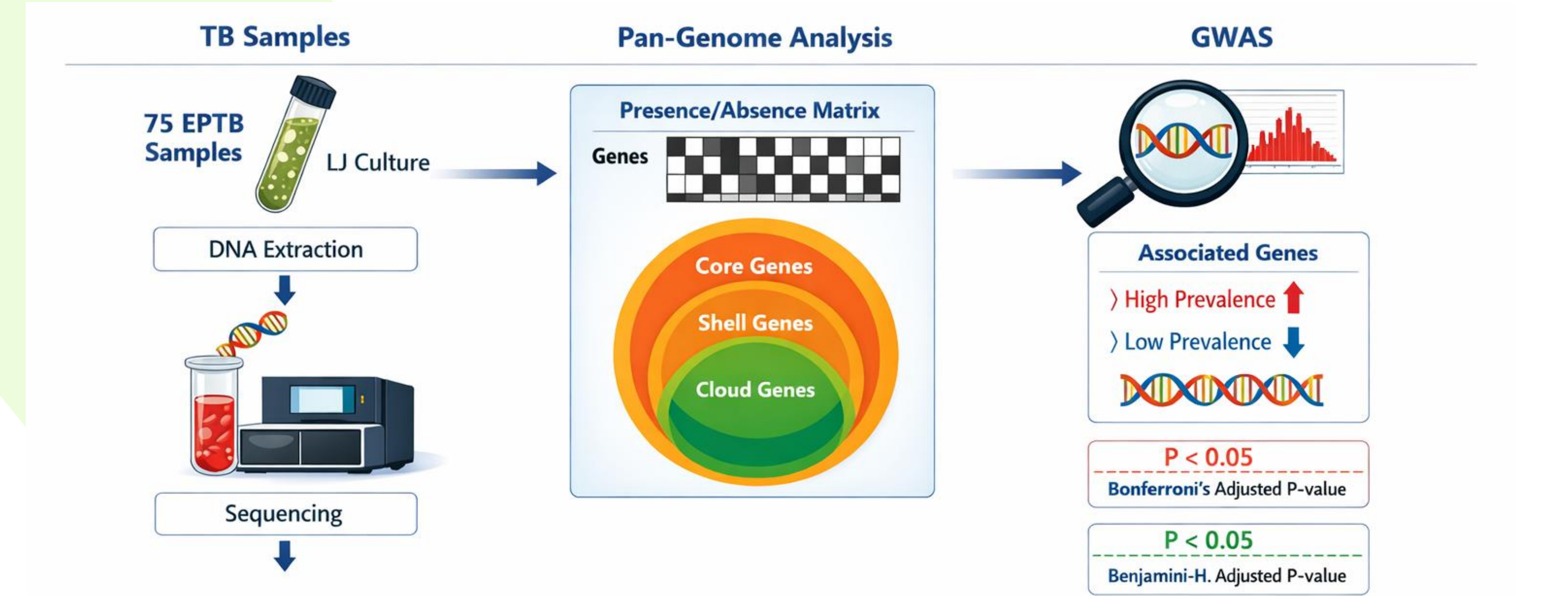
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Introduction

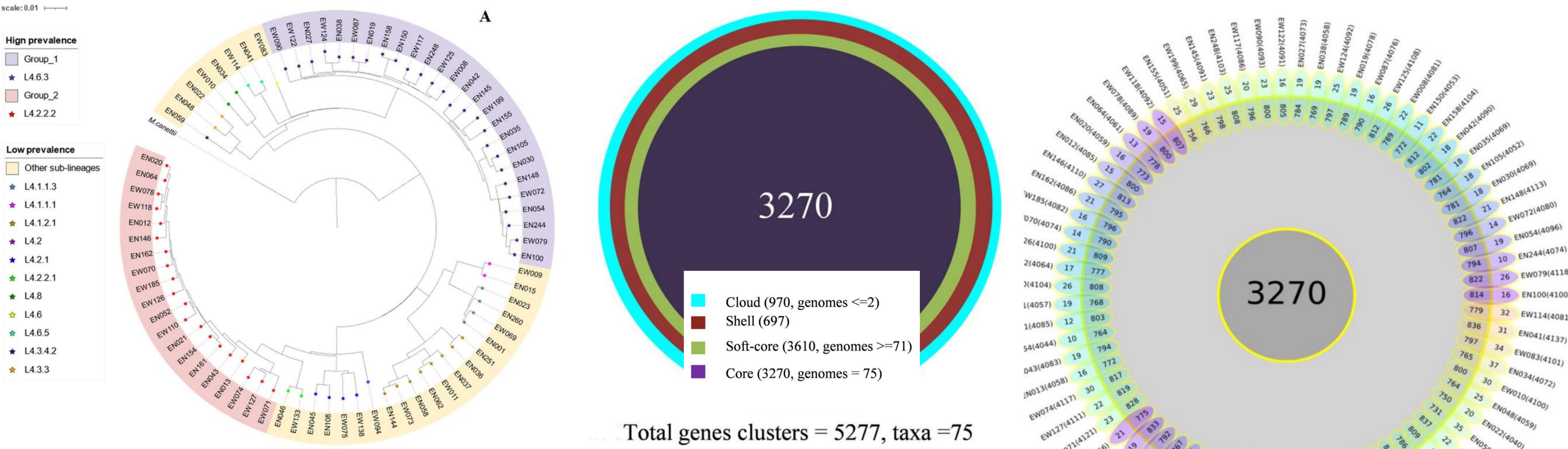
- Tuberculosis (TB) is a major global health problem, still the 1st killer from a single infectious agent.
- Lineage-4 (L4) and its sub-lineages are globally dominant.
- However, certain sub-lineages exhibit higher prevalence than others at the local level.
- In western Ethiopia, for example, L4.2.2.2 and L4.6.3 shows high prevalent.
- The genetic factors underlying the dominance of these highly prevalent Lineages are remain unclear.



Methods



Results and Discussion



Total genes clusters = 5277, taxa =75

When comparing the **high-prevalence L4.6.3** and low-prevalence groups, **11 genes** exhibited significant associations... In a comparison between **high-prevalence L4.2.2.2** and low-prevalence groups, **13 genes** were identified with associations.

Gene	Uniprot code	Length (aa)	Annotation	Genomes number high prev. N = 20	Genomes number low prev. N = 29	P-value	Bonferroni's adjusted P-value ^a	Benjamini-H. adjusted P-value ^a	Genes association
Rv2717c	P9WFG7	164	Hypothetical protein	0	27	8.17E-12	1.35E-08	1.68E-09	LP
Rv0021c	P71591	322	Hypothetical protein	4	29	1.45E-09	2.38E-06	1.99E-07	LP
narX/Rv1736c	P9WJQ1	652	Nitrate reductase-like protein	5	29	9.84E-09	1.62E-05	9.004E-07	LP
arsB1	P9WPD7	428	Arsenic-transport integral membrane protein	0	23	1.63E-08	2.69E-05	1.42E-06	LP
Rv0654	P9WPR5	501	Carotenoid cleavage oxygenase	0	22	3.29E-08	5.42E-05	1.59E-06	LP
Rv0075	O53620	390	Aminotransferase	0	22	3.29E-08	5.42E-05	1.59E-06	LP
Rv0073	P9WQK5	330	Glutamine ABC transporter ATP-binding protein	0	22	3.29E-08	5.42E-05	1.59E-06	LP
Rv1132	O06583	576	Conserved membrane protein	0	22	3.29E-08	5.42E-05	1.59E-06	LP
eccC4	P9WNA7	1236	ESX-4 secretion system protein	0	22	3.29E-08	5.42E-05	1.59E-06	LP
Rv3047c	I6X642	94	Hypothetical protein	0	22	3.29E-08	5.42E-05	1.59E-06	LP
TB27.3	P9WIR3	261	Conserved protein	0	22	3.29E-08	5.42E-05	1.59E-06	LP
PE_PGRS42	I6XEF1	694	PE-PGRS family protein	0	21	5.04E-07	0.0008	1.85E-05	LP
Rv0025	P9WMA1	120	Hypothetical protein	20	15	0.0002	0.32	0.004	HP

HP high prevalence; LP low prevalence
^a adjusted p-values were calculated using Bonferroni's and Benjamini-Hochberg corrections.

Gene	Annotation	Uniprot code	Length (aa)	Genomes number high prev. N = 26	Genomes number low prev. N = 29	P-value	Bonferroni's adjusted P-value ^a	Benjamini-H. adjusted P-value ^a	Gene association
Rv0071	Maturase	O53616	235	0	29	2.81E-16	5.05E-13	7.22E-14	LP
vapC28/Rv0609	Ribonuclease	P9WF81	133	0	29	2.81E-16	5.05E-13	7.22E-14	LP
Rv3098A	PemK-like protein	V5QRX7	106	0	29	2.81E-16	5.05E-13	7.22E-14	LP
icl2/aceA	Isocitrate lyase 2	Q8VJU4	766	0	24	5.18E-09	9.31E-06	3.58E-07	LP
Rv1928c	Short-chain type dehydrogenase/reductase	P95286	255	26	8	5.18E-09	9.31E-06	3.58E-07	HP
PE_PGRS20	PE-PGRS family protein	P9WTF9	463	26	11	2.51E-07	0.0005	1.46E-05	HP
Rv3093c	Oxidoreductase	O05772	334	26	13	2.46E-06	0.0004	0.0001	HP
fadD34	Fatty acid-CoA ligase	L7N699	562	13	29	7.17E-06	0.01	0.0003	LP
PE_PGRS6	PE-PGRS family protein	L0T3X8	594	13	29	7.17E-06	0.01	0.0003	LP
Rv0025	Hypothetical protein	P9WMA1	120	26	15	2.00E-05	0.05	0.0007	HP
Rv2994	MFS-type transporter	P9WJW7	445	16	29	0.0002	0.32	0.005	LP

HP high prevalence; LP low prevalence.
^a adjusted p-values were calculated using Bonferroni's and Benjamini-Hochberg corrections.

Conclusion

L4 employs distinct virulence factors for successful dissemination in western Ethiopia. Given that the functions of these newly identified genes are not well understood, it is advisable to experimentally validate their roles, particularly in the successful transmission of L4.